
Network Architecture Overview

Topology

Three-tier VPC design per region: public subnets hold the load balancer and NAT gateway; private subnets run EKS node groups; isolated subnets hold Postgres, Redis, and Elasticsearch. Cross-region links carry Postgres replication and S3 storage replication.

Traffic flow

- Inbound: CDN caching → WAF rules → load balancer → API gateway → services.
- East-west: mTLS service mesh; NetworkPolicies default-deny per kubernetes conventions.
- Outbound: NAT gateway with an egress allowlist; vendor private links (see vendor security review) bypass the public internet.

Controls

DNS management, VPC config, and IAM policies are Terraform modules with drift detection. Flow logs feed audit logging; anomalies alert through Prometheus to the oncall rotation.